Dietary improvement: Salt iodization remains the most cost effective way to deliver iodine to both humans and livestock and is credited with eradicating iodine deficiency.

CASE STUDY

Case study 1

Margret is having a swelling in her neck which makes it difficult for her to breathe and swallow food. The enlargement is soft and does not pain.

1. What is the condition she is suffering from?
2. What is the nutrient deficient in her food?
3. Suggest some foods for her quick recovery

(v) Zinc

- **Distribution:** Zinc is the most important intracellular trace element. An adult human contains 2g of zinc of which 60% is in skeletal muscle, 30% in bone and 4-6% is present in zinc.
- **Functions:** The functions of zinc include
  - Important constituent of enzymes like alkaline phosphatase and carbonic anhydrase.
  - Required by protein kinases that participate in gene expression.
  - Also a component of metalloenzymes.
- **Food sources:** Meat, seafood and liver are good sources of bioavailable zinc. In cereals most of the zinc is found in the outer fiber rich part of the kernel.



- **Deficiency:** The clinical manifestations of severe zinc deficiency in humans are growth retardation, dermatitis, hair loss, diarrohea, increased infections, delayed wound healing, loss of appetite, hypoguesia (diminished taste) dysguesia (altered taste) hyposmia (diminished smell). Decreased zinc intake is associated with increased risk of low birth weight and preterm delivery.

5.3.2. Vitamins

Fat soluble vitamins A, D, E and K and also water-soluble vitamins C and B group are found in foods. These are needed for

Fat Soluble	Water Soluble
A, D, E, K	B complex group and C

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